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Solving Linear Equations

System of linear equations

$Ax = b$

$Ax = 0$: trivial soln : $x = \vec{0}$] → only soln if col^ms of A are LI.
non-trivial soln : if col^ms of A are LD.

Row picture : we draw lines / planes and get point of intersec^n.

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col^m picture:

$$x\begin{pmatrix}1\\2\end{pmatrix} + y\begin{pmatrix}-1\\1\end{pmatrix} = \begin{pmatrix}3\\4\end{pmatrix}$$

← linear comb. of columns of $m = \begin{bmatrix}1 & -1\\2 & 1\end{bmatrix}$

$mx = B$

$mx = b$

- soln exists (consistency)
- no soln (inconsistency)

row picture : no intersection point

col^m picture : The col^ms of m and B together are LI.
B can't be represented as lin. comb^n of col^m vectors.
span of col^m vectors doesn't have B.

$x + y + z = 4$
$2x + y + z = 3$ } → $x = -1$

$3x + 2y + 2z = 7$ → $y + z = 5$

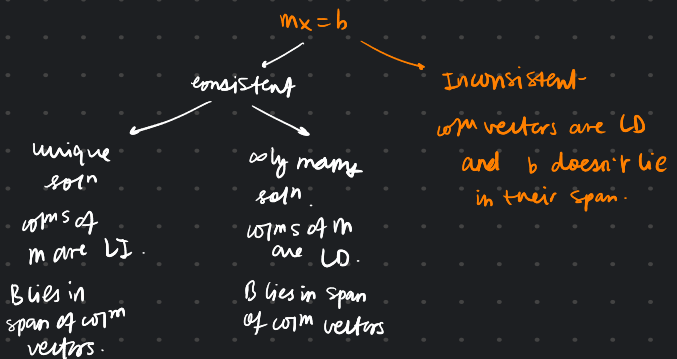
↳ This lies on intersec^n of above planes.

only many soln^s.

col^m picture :

$$\begin{bmatrix}1 & 1 & 1\\2 & 1 & 1\\3 & 2 & 2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}4\\3\\7\end{bmatrix}$$

$\begin{pmatrix}1\\2\\3\end{pmatrix}\begin{pmatrix}1\\1\\2\end{pmatrix}\begin{pmatrix}1\\1\\2\end{pmatrix}$ Do not span $\mathbb{R}^3$.

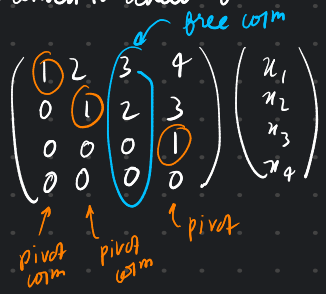


Row echelon form

- All non-zero rows are above zero rows.
- All entries below leading entry (in corresp col^m) must be 0.
- leading entry of any row should be to the right of that of rows above.

Free and pivot variables

- first convert to echelon form.

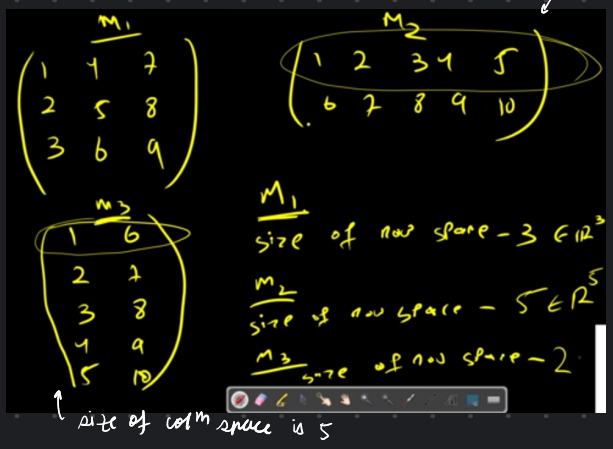


pivot colm: Has leading entry.

$\{x_1 \; x_2 \; x_4\}$: pivot variables

$\{x_3\}$: free var.

A colm cannot have 2 pivots.



Row reduction

Operations:
- row swapping: swap any 2 rows.
- scalar multiplication: multiply any row with a non-zero scalar.
- row addition: add or subtract a multiple of a row to/from another.

E.g. $\begin{bmatrix}1&2&3\\4&5&6\\2&3&4\end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{\text{valid}} \begin{bmatrix}1&2&3\\2&3&4\\4&5&6\end{bmatrix} \xrightarrow{\text{invalid}} \begin{bmatrix}1&2&4\\2&4&3\\4&6&5\end{bmatrix}$ colm swapping is invalid.

Reduc^n to echelon form:

$\begin{bmatrix}4&5&6\\1&2&3\\2&3&4\end{bmatrix} \; R_1 \leftrightarrow R_2$

$\begin{bmatrix}1&2&3\\4&5&6\\2&3&4\end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 4R_1} \begin{bmatrix}1&2&3\\0&-3&-6\\0&-1&-2\end{bmatrix} \xrightarrow[R_3 \leftarrow \frac{R_3}{-1}]{R_2 \leftarrow \frac{R_2}{-3}} \begin{bmatrix}1&2&3\\0&1&2\\0&1&2\end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix}1&2&3\\0&1&2\\0&0&0\end{bmatrix}$

Steps:
- Start with the first row and ensure that the pivot isn't zero. If it is zero, swap with a lower row that has a non-zero value in the same column.
- Make all entries below the pivot zero (by subtracting appropriate multiples of the pivot row from the rows below).
- Move to the next row and repeat:
  Find the pivot in the next row and eliminate all values below it.

Q. $\begin{bmatrix} 2 & 3 & 4 \\ 2 & 5 & 6 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 1R_1} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2/2} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \end{bmatrix}$

Q. $x + 2y + 3z = 9$
$2x + 3y + 5z = 23$
$4x + y + 2z = 15$

$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 4 & 1 & 2 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 4R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & -7 & -10 \end{bmatrix} \xrightarrow[R_3 \leftarrow \frac{R_3}{-1}]{R_2 \leftarrow \frac{R_2}{-1}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 7 & 10 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - 7R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{bmatrix}$

Augmented matrix

$x + 2y + 3z = 9$
$2x + 3y + 5z = 23$ $\Big\}$ $Ax = b$
$4x + y + 2z = 15$

Augmented matrix = $[A|b]$

$[A|b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 2 & 3 & 5 & 23 \\ 4 & 1 & 2 & 15 \end{array}\right]$

we need to check the echelon form criteria for the coefficient part.
In $[A|b]$, A must be in row-echelon form.

$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 2 & 3 & 5 & 23 \\ 4 & 1 & 2 & 15 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 4R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & -1 & -1 & 5 \\ 0 & -7 & -10 & -21 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 7R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & -3 & -56 \end{array}\right] \xrightarrow[R_3 \leftarrow \frac{R_3}{-3}]{R_2 \leftarrow \frac{R_2}{-1}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 1 & 56/3 \end{array}\right]$

$x + 2y + 3z = 9$
$y + z = -5$
$z = \frac{56}{3}$

$y + \frac{56}{3} = -5$
$\Rightarrow y = -\left(5 + \frac{56}{3}\right)$
$y = -\frac{71}{3}$

$x - \frac{142}{3} + 56 = 9$
$x = -\left(47 + \frac{142}{3}\right) = -\frac{283}{3}$

Q.
$2x + 3y - z = 5$
$4x + 4y - 3z = 3$
$2x - 3y + 2z = 1$

$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 4 & 4 & -3 & 3 \\ 2 & -3 & 2 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 0 & -2 & -1 & -7 \\ 0 & -6 & 3 & -4 \end{array}\right] \xrightarrow[R_3 \leftarrow \frac{R_3}{-1}]{R_2 \leftarrow \frac{R_2}{-1}} \left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 0 & 2 & 1 & 7 \\ 0 & 6 & -3 & 4 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 0 & 2 & 1 & 7 \\ 0 & 0 & -6 & -17 \end{array}\right]$

$-6z = -17$
$z = 17/6$
$2y + \frac{17}{6} = 7$
$y = \frac{1}{2}\left(7 - \frac{17}{6}\right) = \frac{1}{2} \cdot \frac{25}{6} = \frac{25}{12}$

$2x + 3y - z = 5$
$2x + \frac{25}{4} - \frac{17}{6} = 5$
$2x = 5 - \frac{25}{4} + \frac{17}{6} = \frac{17}{6} - \frac{5}{4} = \frac{68 - 30}{24} = \frac{19}{12}$

$Ax = b$. Say A is an $m \times n$ coeff matrix.

of variables in system of eqns = n = # of colms

of eqns = m = # of rows.

e.g. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x\begin{bmatrix} 1 \\ 1 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \end{bmatrix} + z\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Q. $A_{m \times n}$ is a coeff matrix. $m < n$. How many soln's does $Ax = 0$ has?

→ # of eqns < # of variables.

∞ly many soln's.

If $m < n$, colm vectors will be LD. So ∞ly many soln's.

$s_1 v_1 + s_2 v_2 + \dots + s_n v_n = 0$

there exists some non-zero s_i that satisfies this, $(v_1, \dots, v_n)$ are LD.

If $(v_1, \dots, v_n)$ is LD, there's some non-trivial soln (non-zero).

Systems of linear eqns with ∞ possible soln's:

Q. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 7 \\ 3 & 4 & 2 & 10 \end{array}\right]$

→ $R_2 \leftarrow R_2 - 2R_1$

$R_3 \leftarrow R_3 - 3R_1$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$x + y + z = 3$

$y - z = 1$

Let $z = t$

$\Rightarrow y = 1 + t$

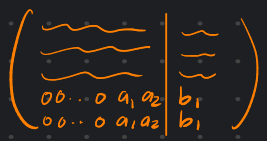
$\Rightarrow x = 3 - (1+t) - t$

$= 2 - 2t$

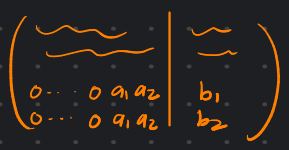
soln is $\begin{pmatrix} 2-2t \\ 1+t \\ t \end{pmatrix}$

or $x\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + z\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$

pivot pivot free



→ ∞ly many soln's.



→ No soln.

Q. Given that coeff. matrix has LI colms. For $Ax = b$, what can we say about the soln space?

→ If for $m \times n$ matrix, $m > n$, e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$x = 0$

$y = 0$ → Inconsistent.

$0 = 1$

so if b lies in span of LI colm vectors, then unique soln.

b doesn't lie in the span of LI colm vectors, then no soln.

If LI vectors → we can't have ∞ly many soln's.

Q. $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 3 & 6 & 3 & 12 \end{array}\right]$

→ $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

2 free, 2 free

$x + 2y + z = 4$

let $z = t$

$y = u$

$x = 4 - 2u - t$

soln: $\begin{pmatrix} 4-2u-t \\ u \\ t \end{pmatrix}$

Q. $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \\ 3 & 6 & 3 & 0 \end{pmatrix}$

→ $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

soln $\begin{pmatrix} -2s-t \\ s \\ t \end{pmatrix}$

soln $= s\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

$s_1v_1 + s_2v_2 + \dots + s_nv_n = 0$

$s_1\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + s_2\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + s_3\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 0$

↑ there are non-zero s_i's.

soln space is span of $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

↑ ↑

These are 2 LI soln's for $Ax = 0$.

$(-2\ 1\ 0)$ and $(-1\ 0\ 1)$

a_1 a_2

a_1, a_2 are LI vectors for the soln space of $Ax = 0$.

$\Rightarrow \{a_1, a_2\}$ forms a basis of the soln space of $Ax = 0$.

n free vars → n basis vectors.

Q. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 3 & 5 & 0 \\ 4 & 5 & 6 & 0 \end{array}\right]$

→ $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$

$x_3 = 0$

$x_2 + 3x_3 = 0$

$\Rightarrow x_2 = 0$

$x_1 + x_2 + x_3 = 0$

$\Rightarrow x_1 = 0$

soln $= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

If cols of A are LI, only the trivial soln is possible.

Trivial soln for $Ax = 0$ → zero vector. $Ax = b$ can't have a trivial soln when $b \neq 0$.

If $Ax = 0$ has unique soln → it'll be trivial. $Ax = 0$ will have 1 soln at least

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If # of pivots = # of free vars ⇒ col vectors are LI.

else LD.
